Euclid's Elements — Book I
Proposition 27

CGJteamLab

If a straight line falling on two straight lines make the alternate angles equal to one another, the straight lines will be parallel to one another.

1 Introduction

Proposition I.27 is the first proposition in Book I whose conclusion is parallelism. Its mathematical content is the converse direction of the familiar alternate-angle criterion:

$$\text{equal alternate interior angles} \quad \Longrightarrow \quad \text{parallel straight lines.}$$

No form of Euclid's Fifth Postulate is required. The proposition belongs to neutral geometry: Euclid proves it by showing that any hypothetical intersection of the two lines would contradict Proposition I.16.

The present formalization is especially interesting because the Hilbert development already contains the alternate-angle implication as a general theorem. At the same time, translating the classical transversal diagram into the hypotheses of that theorem exposes additional incidence, order, ray-orientation, and side-of-line information that is suppressed in the traditional picture.

For this reason the Lean file deliberately retains two formulations of Proposition I.27. Their precise proof-theoretic relation is left visible rather than being eliminated by premature refactoring.

2 Classical Statement

Let a straight line meet the straight lines AB and CD at E and F, respectively. Suppose that the alternate interior angles

$$\angle AEF \qquad \text{and} \qquad \angle EFD$$

are equal. Then

$$AB \parallel CD.$$

The basic configuration is shown in Figure 1.

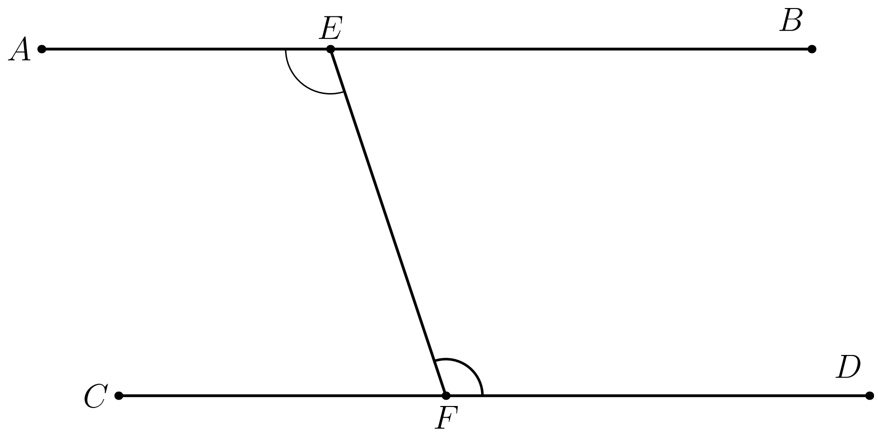

Figure 1: The transversal configuration of Proposition I.27. The alternate interior angles $\angle AEF$ and $\angle EFD$ are congruent.

3 Euclid's Proof

The proof is by contradiction.

Suppose first that the straight lines AB and CD, when produced beyond B and D, meet at a point G. Then $\angle AEF$ is an exterior angle of the triangle EFG. By Proposition I.16,

$$\angle EFD < \angle AEF.$$

This contradicts the hypothesis

$$\angle AEF \cong \angle EFD.$$

The corresponding reductio configuration is shown in Figure 2.

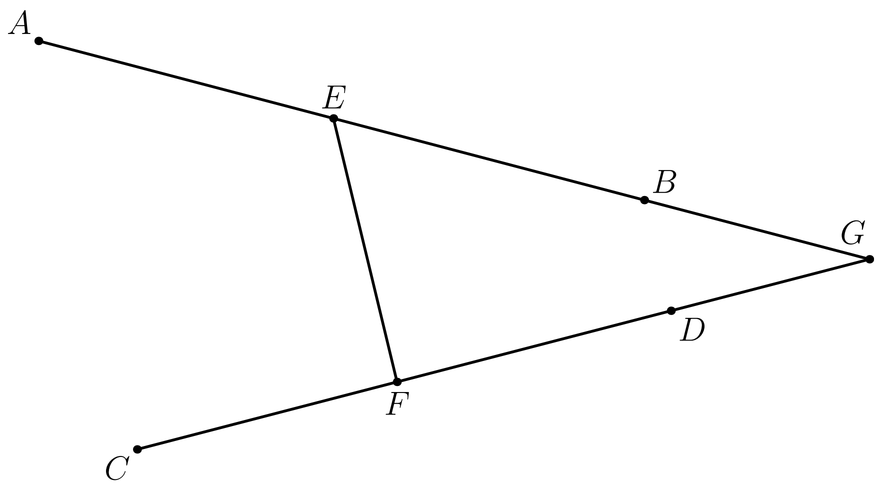

Figure 2: First reductio branch. If the two lines meet beyond B and D at G, Proposition I.16 makes $\angle AEF$ strictly greater than $\angle EFD$.

There is a second possible direction of intersection. Suppose that the lines, when produced beyond A and C, meet at a point H. Then $\angle EFD$ is an exterior angle of the triangle EFH. Again Proposition I.16 gives

$$\angle AEF < \angle EFD,$$

which contradicts the same angle-congruence hypothesis.

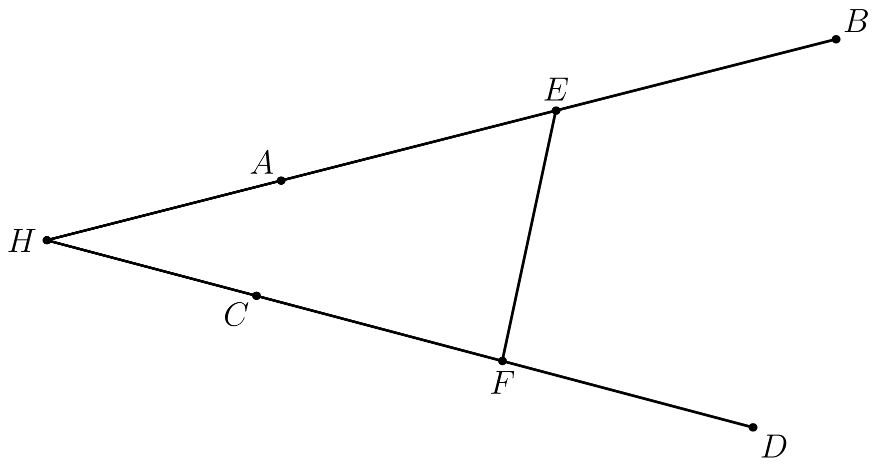

Figure 3: Second reductio branch. If the two lines meet beyond A and C at H, Proposition I.16 makes $\angle EFD$ strictly greater than $\angle AEF$.

Both possible directions of intersection therefore lead to contradictions. Hence the two straight lines do not meet:

$$AB \parallel CD.$$

This is exactly the classical content of Proposition I.27.

4 Diagrammatic Audit

Three figures are retained because they represent three logically distinct configurations.

Figure 1 records only the hypothesis of the proposition: a transversal and a pair of congruent alternate interior angles.

Figure 2 represents the first hypothetical intersection, beyond B and D. Figure 3 represents the second hypothetical intersection, beyond A and C.

The last two figures are therefore not construction stages and are not simultaneously realizable with the original angle-congruence hypothesis. They are diagrams of the two reductio branches. Their purpose is to make the two applications of Proposition I.16 geometrically transparent.

As in Proposition I.26, the figures should encode the local geometric step being used rather than accumulate every relation occurring elsewhere in the proof.

5 Hilbert Reconstruction

The Hilbert development reaches the mathematical core of I.27 before the Euclidean sequence reaches Proposition I.27.

At the synthetic-interface level the implication

$$\text{equal alternate angles} \quad \Longrightarrow \quad \text{parallel lines}$$

is already available through

```
parallel_from_equal_angles
```

The first Lean formulation therefore exposes I.27 directly at this level. Its proof is only an application of the existing theorem.

This is not an appeal to the Euclidean parallel postulate. The alternate-angle implication is a theorem of the neutral Hilbert development. The logical direction is important: I.27 proves a sufficient condition for parallelism; the converse direction used in Euclidean geometry appears later, in Proposition I.29, where the Fifth Postulate enters.

6 Two Formal Formulations

The current Lean file intentionally retains two formulations.

6.1 The interface-level formulation

The theorem

```
euclid_proposition_27
```

uses the ordered configuration already expected by `parallel_from_equal_angles`. Its proof is:

```
exact
  parallel_from_equal_angles
    Geo
    A C D
    B E F
    hADC
    hCEB
    hDEF
    hCED
    hAngle
```

At this level the central mathematical content of Proposition I.27 has already been packaged by the Hilbert theory. The Book I theorem merely exposes that result in the Euclidean sequence.

6.2 The explicit transversal formulation

The second theorem is

```
euclid_proposition_27_transversal
```

and begins with a configuration closer to the classical diagram:

$$A - E - B, \qquad C - F - D,$$

with E and F on an explicitly supplied transversal *trans*. It also records that the selected interior arms EA and FD lie on opposite sides of the transversal and assumes

$$\angle AEF \cong \angle EFD.$$

This version reveals information that the traditional phrase "alternate interior angles" leaves implicit.

7 Normalization of the Transversal Configuration

The explicit proof first chooses a point M between E and F:

$$E - M - F.$$

This provides explicit representatives for the two directions of the transversal. Betweenness gives the corresponding same-ray relations. After reversing the first angle and transporting its transversal arm along these rays, the original congruence

$$\angle AEF \cong \angle EFD$$

is normalized to

$$\angle MEA \cong \angle MFD.$$

The theorem

```
hilbert_parallel_of_alternate_angles_oppositeSide_lines
```

then yields the local parallelism

$$EA \parallel FD.$$

The remainder of the proof is transport along the two original straight lines. Since

$$A - E - B$$

the first local line is extended from EA to AB. Since

$$C - F - D,$$

the second is transported from FD to CD. Thus

$$AB \parallel CD.$$

8 Proof Architecture

The classical and formal routes may be displayed schematically as follows.

```
Euclid
------
angle AEF ~= angle EFD
        |
assume AB and CD meet
        |
        +-----------------------------+
        |                             |
 meet beyond B,D                meet beyond A,C
        |                             |
 exterior angle AEF             exterior angle EFD
        |                             |
       I.16                          I.16
        |                             |
 EFD < AEF                      AEF < EFD
        |                             |
 contradiction                   contradiction
        +-------------+---------------+
                      |
                   AB || CD


Hilbert interface
-----------------
ordered alternate-angle configuration
        |
parallel_from_equal_angles
        |
     parallel


Explicit transversal formalization
----------------------------------
A-E-B, C-F-D
+ transversal EF
+ opposite-side information
+ angle AEF ~= angle EFD
        |
choose M with E-M-F
        |
same-ray and angle normalization
```

```
        |
angle MEA ~= angle MFD
        |
hilbert_parallel_of_alternate_angles_oppositeSide_lines
        |
EA || FD
        |
transport along A-E-B and C-F-D
        |
AB || CD
```

9 Proof-Theoretic Status

The coexistence of the two Lean formulations is deliberate.

The short formulation makes the mathematical core of I.27 unusually visible: the proposition coincides closely with an already established general theorem of the Hilbert layer.

The explicit transversal formulation answers a different question: what formal information is required to pass from the classical diagram to that general theorem? Its additional work concerns incidence, betweenness, same rays, side-of-line data, orientation of angles, and transport of parallelism.

At present we do not force a final interpretation of this distinction. It may eventually be preferable to regard the short theorem as the canonical Book I interface and the transversal theorem as a normalization result. It may instead prove useful to retain the explicit configuration as part of the public Euclidean layer.

Keeping both versions preserves this question for later comparison with the proof corpus and with the dependency graph of the Hilbert reconstruction.

10 Relationship with Proposition I.16

Euclid's classical proof depends directly on Proposition I.16. Each possible intersection converts one of the supposedly equal alternate angles into an exterior angle of a triangle, and I.16 makes that angle strictly greater than the remote interior angle.

The present Lean theorem does not invoke `euclid_proposition_16` directly at the Book I level because the corresponding alternate-angle theorem has already been established inside the Hilbert theory.

Thus the difference is one of proof organization:

$$\begin{aligned} \text{Euclid} &: \text{I.16} \longrightarrow \text{I.27}, \\ \text{Hilbert reconstruction} &: \text{order and angle theory} \longrightarrow \text{alternate-angle parallelism} \longrightarrow \text{I.27}. \end{aligned}$$

The Book I order and the formal dependency order therefore need not coincide.

11 Neutral Geometry and the Fifth Postulate

Proposition I.27 is a useful logical boundary marker.

It establishes

$$\text{equal alternate interior angles} \implies \text{parallel lines}$$

without using the Fifth Postulate. This direction is valid in neutral geometry.

The converse statement,

$$\text{parallel lines} \implies \text{equal alternate interior angles},$$

is part of Proposition I.29 and requires Euclidean parallelism.

Consequently I.27 should not be classified as a theorem depending on Hilbert's Euclidean parallel axiom merely because its conclusion contains the word "parallel". Its proof uses the definition of parallelism as nonintersection, together with the neutral theory developed before it.

12 Lean Formalization

The complete implementation is contained in `Proposition27.lean`. The public file currently records both the interface-level theorem and the explicit transversal theorem.

The latter proof has the following principal stages:

1. choose M with $E - M - F$;
2. obtain the two same-ray relations along the transversal;
3. normalize the given angle congruence;
4. apply the Hilbert alternate-angle parallelism theorem;
5. transport the resulting local parallelism along the collinear triples $A - E - B$ and $C - F - D$.

No Euclidean parallel axiom is used.

13 Conclusion

Proposition I.27 is formally small at the level of the developed Hilbert interface but conceptually revealing.

Euclid proves parallelism by excluding the two possible intersection directions through Proposition I.16. The Hilbert development has already abstracted this reasoning into a reusable alternate-angle theorem, so the core Book I formulation becomes a thin wrapper.

The explicit transversal theorem shows what that compression hides: the classical diagram silently carries order, orientation, side-of-line, and ray information that must be made explicit before the abstract theorem can be applied.

For the moment both formalizations are retained. Their coexistence is not duplication to be removed immediately, but a useful record of the difference between the mathematical content of I.27 and the formal normalization of its classical diagram.